Second-Moment Itô Mass-Flux Tracers in AthenaK

Implementation Decisions, Validation, and Thermodynamic Tracing

Technical note for `feature/ito_tracers`

Corrected implementation: `feature/ito_tracers`, June 7, 2026

Abstract

The goal was to add the second-order Itô tracer method to AthenaK as a continuous-path alternative to the existing Monte Carlo mass-flux tracers, while explicitly excluding the third-order method. The implementation derives cell-centered mean displacements and variances from the same final Runge–Kutta-weighted mass fluxes used by the Monte Carlo tracer. A runtime option selects either the independent-coordinate covariance used by the published code or the full finite-step covariance tensor. The coefficient fields are communicated across MeshBlocks and refinement boundaries, interpolated to each particle, and used for the stochastic step.

The evidence supports a precise conclusion: the opt-in full-covariance Itô-2 particles match the vector mean and central covariance of the finite-step Monte Carlo kernel in controlled two- and three-dimensional tests. The default published mode matches the same means and marginal variances. Both preserve continuous subcell positions through AMR motion and provide useful mass-weighted temperature histories in a two-phase thermal-instability calculation. The evidence does not establish a third-order method, the full Monte Carlo probability distribution, a converged thermal-fragmentation solution, or support outside the explicitly validated non-relativistic, periodic, two- and three-dimensional configurations.

1 What We Were Trying to Accomplish

Monte Carlo mass-flux tracers are tied directly to finite-volume mass exchange: particles jump between cell centers with probabilities derived from the mass fluxes. That makes their ensemble transport physically aligned with the discrete fluid update, but individual paths are discontinuous and cell-centered. For thermodynamic tracing, those jumps can obscure the continuous history one would like to associate with a moving mass element.

The requested feature was therefore narrow:

1. build on the existing Monte Carlo tracer method rather than introduce an unrelated passive-particle model;
2. implement only the second-order Itô method from the method described by Moseley, Teyssier, and Abel (arXiv:2604.23041v2);
3. preserve the published independent-coordinate method for compatibility, while making full finite-step covariance available when needed; and
4. verify that the particles can record meaningful thermodynamic histories in a coupled cooling problem.

2 The Simple Picture

For one cell, the Monte Carlo tracer chooses one face jump or stays put. The Itô-2 tracer replaces that discrete draw with a continuous random displacement. Both runtime modes match the vector mean and marginal variances. The full mode also matches the cross covariance.

If p_R and p_L are the outward Monte Carlo jump probabilities, define

$$C_+ = p_R + p_L, \qquad C_- = p_R - p_L. \tag{1}$$

For cell width h_i, the finite-step mean and raw second moment are

$$m_i = h_i C_{-,i}, \qquad R_{ij} = h_i^2 C_{+,i} \delta_{ij}. \tag{2}$$

The central covariance is

$$Q_{ij} = R_{ij} - m_i m_j = h_i^2 C_{+,i} \delta_{ij} - h_i h_j C_{-,i} C_{-,j}. \tag{3}$$

Thus the off-diagonal covariance is generally nonzero even though the raw MC cross moment vanishes. With `ito_covariance_model=full_finite_step`, AthenaK factors $\mathbf{Q} = \mathbf{L}\mathbf{L}^\mathsf{T}$. With the default `published_diagonal`, it keeps only $Q_{ij}^{\text{pub}} = \delta_{ij} Q_{ii}$. In either case it advances

$$\mathbf{X}^{n+1} = \mathbf{X}^n + \mathbf{m} + \mathbf{L}\boldsymbol{\xi}, \tag{4}$$

where the components of $\boldsymbol{\xi}$ are independent bounded uniform variables with zero mean and unit variance. The full model correlates the physical coordinate kicks through $\mathbf{L}$; the published model uses a diagonal factor.

The bounded-uniform base law is not rotationally invariant. Changing $\mathbf{L}$ therefore changes fourth and higher moments too, not just the cross covariance. The full mode matches the Monte Carlo mean and covariance, but not its complete categorical jump distribution. A two-mode comparison is thus a comparison of the complete kick laws, not a clean ablation of one tensor entry.

This is the part that matters: the particles still take their transport statistics from the finite-volume mass flux, but they no longer have to sit on cell centers or move by discrete one-cell jumps.

3 Design Choices and Tradeoffs

The thermal-instability problem followed the same conservative approach. It uses the existing `cooling_test` problem generator, the standalone ISM cooling/heating module, and the existing initial-perturbation module. Density-only 1% fluctuations were chosen instead of temperature or pressure noise because they preserve a uniform initial temperature while cleanly triggering the density dependence of cooling.

4 What Was Implemented

The new Itô path is integrated into the existing Lagrangian mass-tracer system rather than implemented as a separate particle subsystem.

1. Hydro and MHD accumulate mass flux using the weights that contribute to the final RK1, RK2, or RK3 state. A simple arithmetic average of stage fluxes would not represent the final finite-volume update.

Approach	Strongest advantage	Main weakness	Decision
Keep only discrete Monte Carlo tracers	Already follows the finite-volume mass exchange and has mature seeding, migration, restart, and output paths.	Individual histories remain cell-centered and discontinuous.	Retained as the baseline.
Use ordinary velocity-following passive particles	Simpler continuous trajectories.	Does not derive transport from the numerical mass flux and therefore does not preserve the Monte Carlo tracer's mixing statistics.	Rejected.
Build Itô-2 on the Monte Carlo infrastructure	Continuous paths tied to the mass-flux jump kernel; selectable published marginal-moment or full finite-step covariance behavior.	Requires new coefficient fields, MeshBlock/AMR communication, and inherits limits of the existing migration path.	Implemented.
Implement Itô-3	Intended to match additional information beyond second moments.	Explicitly outside the requested scope because of unresolved correctness concerns.	Rejected and fail-closed.

Table 1: The implementation choices were driven by the requirement that continuous particles remain tied to the finite-volume mass transport.

2. After the fluid integrator, each cell converts its outward mass-flux probabilities into the finite-step mean $\mathbf{m}$, diagonal variances, and, in full mode, the three cross covariances. Invalid probabilities, non-positive-semidefinite covariance beyond tolerance, and non-finite values terminate the run.
3. The mean and central-covariance fields are restricted, communicated, and prolonged across MeshBlocks and AMR boundaries.
4. Cloud-in-cell interpolation evaluates $\mathbf{m}$ and $\mathbf{Q}$ directly at each particle position.
5. Full mode uses a fixed-size pivoted semidefinite factorization; published mode takes three scalar square roots.
6. A deterministic hash of tracer tag, cycle, random seed, and coordinate produces the bounded uniform vector used in $\Delta\mathbf{X} = \mathbf{m} + \mathbf{L}\boldsymbol{\xi}$.
7. The existing mass-tracer seeding, migration, restart, VTK output, and thermodynamic-history paths handle the resulting particles.

The implementation also makes its scope explicit. Inputs requesting `pusher=ito3`, an `ito_order` other than 2, or a kick distribution other than the bounded uniform Itô-2 draw fail immediately. The current path requires non-relativistic Cartesian Hydro or MHD, dynamic RK1–RK3 evolution, periodic boundaries in every active direction, and a two- or three-dimensional mesh.

The covariance parameter is `particles/ito_covariance_model`. Allowed values are `published_diagonal` and `full_finite_step`; the former is the default. Ito restart format version 4 stores this choice and rejects a restart under a different model. Legacy version 2 and 3 Ito files automatically select `published_diagonal` and `full_finite_step`, respectively.

5 Evidence from Controlled Transport Tests

5.1 One-dimensional-style moment test

Because the particle module requires a two- or three-dimensional mesh, the one-dimensional-style test confines motion to x_1 on a 128×4 sheet. It starts 4096 tracers in one x_1 cell and follows them for 64

RK2 steps.

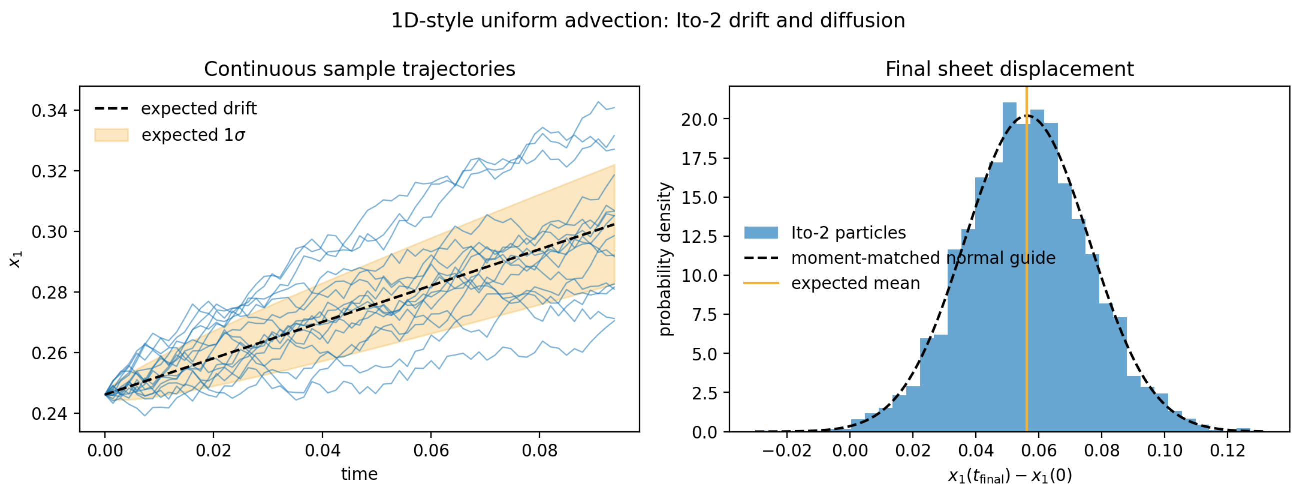

Figure 1: Committed one-dimensional-style behavior test generated by `plot_ito_tracer_figures.py`. The paths are continuous, while the final distribution follows the expected Monte Carlo mean and variance. The measured final mean displacement is 0.05616, compared with the expected 0.05625. This supports per-coordinate moment matching; it does not claim that the continuous distribution reproduces every higher moment of the discrete jump process.

The CPU regression computes the expected mean $\langle \Delta x \rangle = v\Delta t$ and variance $h^2 C(1 - C)$ for uniform advection, then checks 4096-particle samples against those values. It also checks restart behavior and verifies that Itô-3 and invalid particle-type combinations are rejected.

5.2 Two-dimensional continuous paths

The two-dimensional cloud uses 4096 tracers on a 64×64 mesh with diagonal velocity $(v_1, v_2) = (0.45, 0.25)$. Separate MPI/AMR regression coverage checks that particles migrate across ranks, preserve continuous subcell positions through refinement boundaries, and restart from per-rank restart files.

6 Thermodynamic Tracing of Thermal Instability

6.1 Physical setup

The more informative integration experiment combines the Itô-2 particles with ISM-like cooling and initial perturbations in a periodic $20\,\mathrm{pc} \times 20\,\mathrm{pc}$, 128^2 Hydro calculation. The gas starts in exact heating-cooling balance at

$$\frac{P}{k_B} = 3000\ \mathrm{K\,cm^{-3}}, \qquad n_0 = 1.9071755\ \mathrm{cm^{-3}}, \qquad T_0 = 1573.0068\ \mathrm{K}. \tag{5}$$

This is the unstable middle equilibrium. At the initial pressure, the stable equilibria are near 52 K and 6353 K.

The causal chain is simple. The 1% density-only perturbations leave the initial temperature uniform. Slightly denser regions cool faster because radiative cooling scales as n^2, while constant per-particle heating scales as n. They lose pressure and condense. Slightly underdense regions net heat and expand.

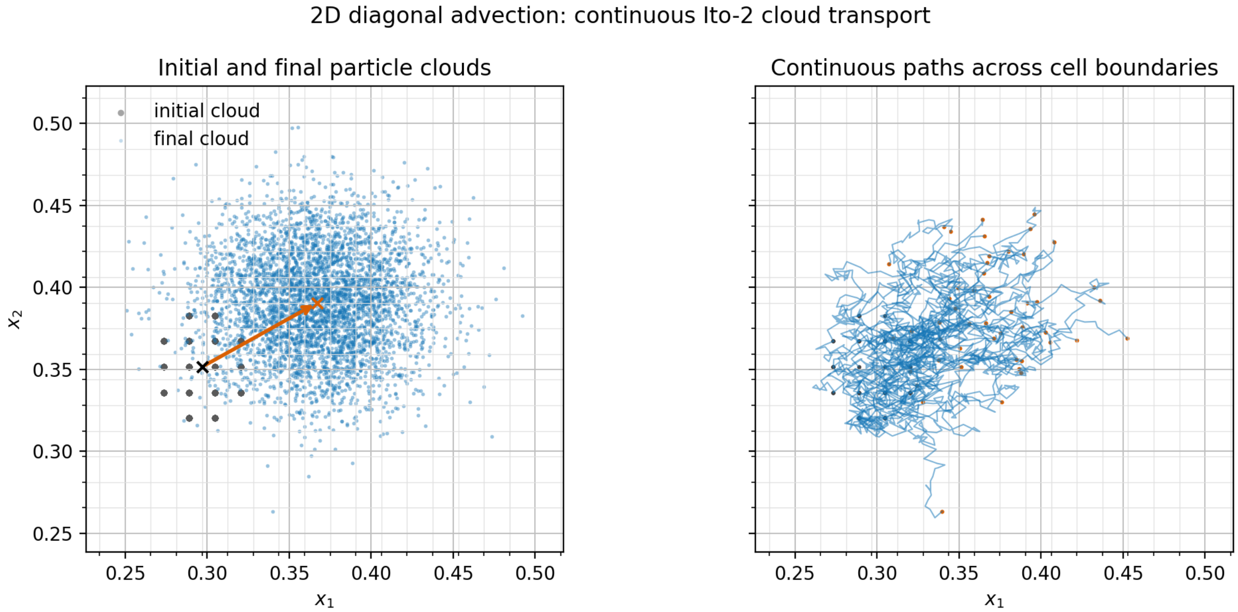

Figure 2: Committed two-dimensional cloud test generated by `plot_ito_tracer_figures.py`. The final cloud both advects and spreads, and the sample paths remain continuous rather than snapping between cell centers. This is direct evidence for the intended geometric behavior, but it is not by itself a test of thermodynamic sampling or AMR communication.

The 4096 mass-weighted Itô tracers record which thermal phase their stochastic mass-flux trajectories occupy over time.

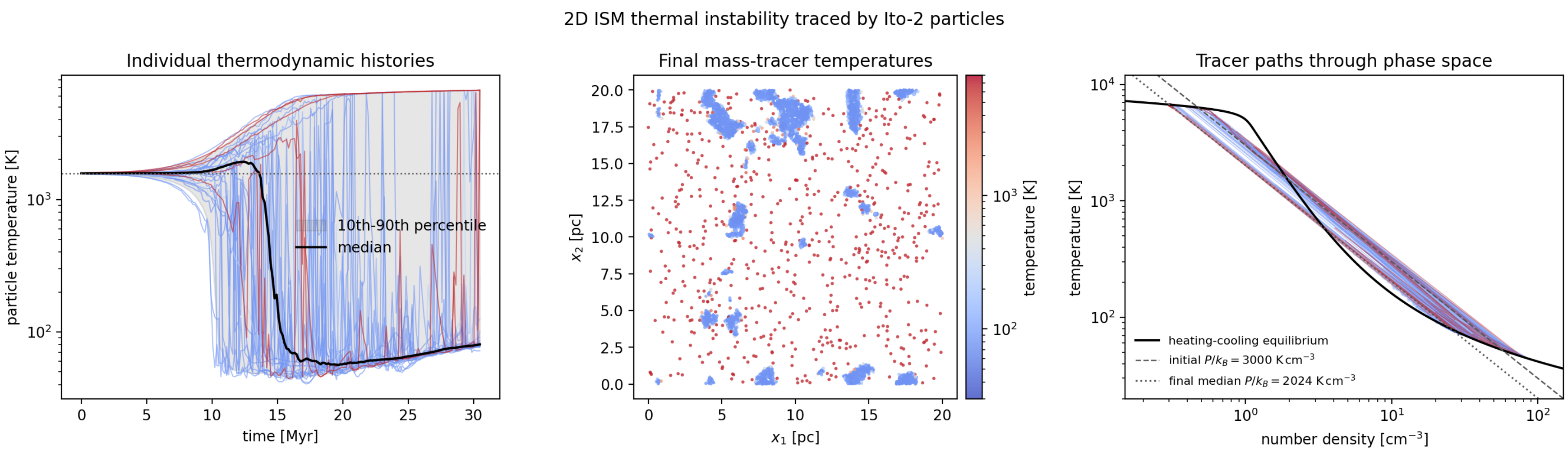

Figure 3: Thermodynamic tracing in the two-dimensional thermal-instability experiment, generated by `plot_ito_thermal_instability.py`. Look first at the left panel: the initially single-temperature population separates into cold and warm histories. The middle panel shows the final mass-tracer temperatures. In the right panel, paths approach the two stable intersections of the heating-cooling equilibrium curve with the evolved median isobar. Sharp jumps in individual histories occur when a particle crosses a thin phase interface because history output samples the fluid cell containing the particle. The figure supports coupled thermodynamic tracing; it does not establish resolution-converged interface structure or fragmentation.

6.2 Observed result

The run completed 100 code time units, or 30.47 Myr, in 9437 cycles. Its thermodynamic-history output contained 823,296 finite records over 201 output cycles for all 4096 tags. At the final output:

- the mass-weighted median pressure was $P/k_B = 2024\ \mathrm{K\,cm^{-3}}$;
- the cooling curve's stable roots at that pressure were 79.5 K and 6720 K, separated by an unstable root at 589 K;
- the measured median tracer temperatures were 79.5 K in the cold phase and 6673 K in the warm phase; and
- the mass-tracer fractions were 81.8% below 300 K, 15.5% above 5000 K, and 2.7% between those thresholds.

The agreement between the measured phase temperatures and the stable heating-cooling intersections is the strongest evidence that the particles are recording the expected thermodynamic evolution. Because the particles were seeded by mass, the reported fractions estimate mass fractions, not volume fractions.

7 Verification and Failure Boundaries

The fail-closed checks are part of the implementation, not just the tests. Runs terminate on invalid or non-finite Monte Carlo probabilities, negative diffusion beyond tolerance, invalid interpolated coefficients, non-finite displacements, and displacements that would cross more than one MeshBlock in an active direction.

8 What the Work Supports, and What It Does Not

What is directly supported.

- Itô-2 is integrated with the existing Monte Carlo mass-tracer infrastructure and final finite-volume mass fluxes.
- Controlled tests support published marginal-moment matching by default and full finite-step vector covariance matching by opt-in, plus continuous-path behavior.
- The particles can record interpretable mass-weighted thermodynamic histories in a coupled two-phase cooling calculation.

What remains uncertain or explicitly unsupported.

- No Itô-3 or third-moment method is implemented or validated.
- Correlated second-moment kicks do not reproduce the full discrete jump distribution or all higher moments; Itô-3 remains unimplemented.
- Current support is limited to non-relativistic Cartesian Hydro/MHD, periodic active boundaries, dynamic RK1–RK3 evolution, and 2D/3D particle meshes.
- The existing particle migration path limits a realized displacement to less than one MeshBlock per coordinate per fluid timestep.
- The initial p_{target}/d timestep guard is not a proof for arbitrary divergent flow. Production-like cases require a full-duration CFL qualification and still fail closed if the realized outgoing probability exceeds one.
- AMR transfers $\mathbf{m}$ and $\mathbf{Q}$ as interpolated stochastic coefficient fields. It does not add the between-child variance required for the covariance of an unresolved mixture of child kernels.
- CUDA/HIP runtime behavior was not tested.

Check	Evidence	Status
Uniform-flow moments	CPU regression compares 4096-particle mean and variance with the analytical Monte Carlo moments; includes RK2 and RK3.	**Passed**
Restart and invalid requests	Serial restart passes; Itô-3, order 3, and wrong particle-type combinations are explicitly rejected.	**Passed**
Full covariance and precision	Opt-in two- and three-dimensional covariance tests include mixed signs, rank-deficient tensors, near-limit probabilities, and single precision.	**Passed**
Published default and restart model	A missing parameter selects the published diagonal law; restart round trips preserve the mode and a mismatched mode is rejected.	**Passed**
Merged cooling and perturbation integration	Combined serial build and 43 focused Itô, cooling, and initial-perturbation tests.	**Passed**
Thermal-instability experiment	Full 128^2, 4096-particle run completed with all history values finite and phase temperatures consistent with stable equilibria.	**Passed**
MPI, AMR, restart, and boundaries	Two- and four-rank decomposition checks, exact MPI/AMR restart equivalence, conditional coarse/fine moments, and exact upper-face/corner wrapping.	**Passed**
Old-versus-corrected turbulence	Three 32^3 paired seeds and one 64^3, 4,194,304-particle pair at empirically qualified CFL 0.324. Gas density is bit-identical.	**Passed; modest scale-dependent changes**
Same-executable runtime	At 64^3 and CFL 0.324, two runs per mode averaged 25.24 s for published diagonal and 37.21 s for full finite-step after hot-path factorization optimization.	**Full mode 1.47x**
GPU runtime	CUDA/HIP execution was not available in this validation environment.	**Not tested; no support claim**

Table 2: Verification status for the corrected implementation.

- The thermal-instability experiment omits conduction, so the Field length is unresolved. Interface thickness, small condensations, and phase fractions are resolution-, seed-, domain-, and runtime-dependent.
- The final particle map samples the mass distribution; it is not a complete volume-filling gas-temperature map.

Bottom line. AthenaK now has a deliberately scoped Itô-2 mass-flux tracer that converts the existing Monte Carlo jump statistics into continuous stochastic paths without disconnecting the particles from the finite-volume mass transport. The thermodynamic-tracing use case works, while third-order behavior, unrestricted configurations, full joint-distribution matching, and converged thermal fragmentation remain outside the supported claim.

9 Reproduction and Artifact Map

The thermal-instability experiment and figure can be reproduced from the repository root with:

```
./build/src/athena \
  -i inputs/particles/ito_tracers_thermal_instability_2d.athinput \
  -d run_ito_ti_2d

python scripts/plot_ito_thermal_instability.py \
  run_ito_ti_2d/prtcl_thermo_history/\
ito_tracers_thermal_instability_2d.prtcl_thermo_history.thp
```

Implementation: `particles_lagrangian_ito.cpp`, `driver.cpp`, the Hydro/MHD flux files, and `particles_tasks.cpp`.

Regression evidence: `test_particles_ito_cpu.py` and `test_particles_ito_mpicpu.py`.

Thermal experiment: `inputs/particles/ito_tracers_thermal_instability_2d.athinput`. Figure script: `scripts/plot_ito_thermal_instability.py`.

User documentation: `docs/source/modules/ito_tracers.md`.